

IBM Quantum

Quantum Leadership Training Program 2026

Pre-work assignment

Welcome to the Quantum Leadership Training Program 2026 sponsored by the Ministry of Science ICT and IBM Quantum! Prior to the **on-site training which begins on November 2 at IBM Korea**, **please start learning and applying quantum computing with Qiskit** by going through this pre-work assignment.

Week 1 Pre-work assignment (5 mandatory + 1 optional)

Foundations of quantum computing

Learn about quantum information from basics to fundamental quantum algorithms

- [Basics of quantum information](#)
- [Fundamentals of quantum algorithms](#)

Key Techniques and applications

Learn about today's quantum computers and how to use them to their full potential.

- [Variational algorithm design](#)
- [Quantum computing in practice](#)
- [Quantum diagonalization algorithms](#)

Optional (choose one)

Explore advanced quantum computing topics, ranging from practical algorithms and applications to foundational theories.

- [Quantum machine learning](#)
- [General formulation of quantum information](#)
- [Foundations of quantum error correction](#)